was the raw pixels on the screen and the idea that it had to get a high score. And everything else it had to figure out by itself.' The program managed to learn the rules of all the games it was presented with, from *Pac-Man* and *Space Invaders* to car racing and tennis games. It then played most of them as well as or better than humans, sometimes coming up with strategies that never occur to human players.[13]



Credit 1.45

45. Deep Blue defeating Garry Kasparov.

Computer algorithms have recently proven their worth in ball games, too. For many decades, baseball teams used the wisdom, experience and gut instincts of professional scouts and managers to pick players. The best players fetched millions of dollars, and naturally enough the rich teams got the cream of the market, whereas poorer teams had to settle for the scraps. In 2002 Billy Beane, the manager of the low-budget Oakland Athletics, decided to beat the system. He relied on an arcane computer algorithm developed by economists and computer geeks to create a winning team from players that human scouts overlooked or undervalued. The old-timers were incensed by Beane's algorithm transgressing into the hallowed halls of baseball. They said that picking baseball